Inequities in Health Care Services Caused by the Adoption of Digital Health Technologies: Scoping Review

Automated LLM-assisted Review Pipeline

Abstract

This scoping review aims to identify and describe inequities in health care services caused by the adoption of digital health technologies. We systematically searched and retrieved 5,736 records, of which 3,395 were ranked after DOI metadata cleaning. From the top 1,000 ranked pool, 200 article cards were generated, and 135 studies were included after screening. Based on an evidence-card completeness and extraction audit, we synthesized findings across eight themes: Sociodemographic Disparities in Access and Use (93 articles), Infrastructure and Technological Barriers (66), Policy, Governance, and Scalability (48), Digital and Health Literacy Barriers (39), Inequities in Disease Outcomes and Care Quality (34), Trust, Privacy, and Security Concerns (31), Equity-Focused Design and Assessment Frameworks (19), and Usability and User-Centered Design (19). Studies could belong to multiple themes, so theme counts exceed the number of included studies. The review reveals that digital health technologies risk exacerbating existing health inequities through complex interactions of sociodemographic factors, literacy barriers, inadequate infrastructure, and lack of user-centered design Reiners et al. [2019], Ahmed et al. [2020], Turnbull et al. [2020], Källander et al. [2013]. While equity-focused frameworks and policies are emerging, evidence gaps remain in scalable interventions, long-term outcomes, and trust-building strategies. These findings underscore the need for intentional equity integration across the digital health lifecycle.

1 Introduction

1.1 Background

Digital health technologies—ranging from telemedicine and mobile health applications to electronic health records and wearable devices—have been widely promoted as tools to improve access, efficiency, and quality of health care services. Their rapid adoption, particularly in the last two decades, has been driven by expectations of reducing costs, enhancing patient engagement, and enabling remote care. However, the same technologies that promise to bridge gaps in health service delivery can also introduce or deepen existing disparities. Socially disadvantaged groups—including those with low income, limited digital literacy, older adults, rural populations, and ethnic minorities—may face barriers to accessing and effectively using these innovations. The phenomenon often referred to as the digital divide has been documented across multiple contexts, yet a comprehensive understanding of how digital health technologies themselves contribute to inequities in health care services remains fragmented.

1.2 Motivation and gap

While numerous primary studies and some reviews have examined equity aspects of digital health, the literature tends to focus on either access barriers or outcomes in specific populations, without synthesizing the full range of inequities that arise from the adoption of digital health technologies

across different health care settings. Existing reviews often concentrate on a single technology type, disease area, or geographic region, limiting their ability to reveal cross-cutting patterns and mechanisms. Moreover, the methods used to assess equity vary widely, making it difficult to compare findings or derive actionable insights. A scoping review that systematically maps the evidence on inequities caused by the adoption of digital health technologies is therefore needed to clarify the scope of the problem, identify common themes and evidence gaps, and inform future research and policy.

1.3 Review objective

The objective of this scoping review is to identify and describe the inequities of health care services brought about by the adoption of digital health technologies.

1.4 Research questions

The review is guided by the following research questions:

- What types of inequities in health care services have been associated with the adoption of digital health technologies?
- Which population groups are most affected by these inequities, and in which health care contexts?
- How have these inequities been conceptualized, measured, or reported in the literature?
- What patterns, mechanisms, and gaps emerge from the existing evidence?

1.5 Contributions

This scoping review provides a broad synthesis of the literature on inequities related to digital health adoption, drawing from multiple databases and spanning the period from 1990 to 2020. It identifies recurrent themes, contrasts different methodological approaches, and highlights persistent evidence gaps. The findings are intended to support researchers, health care providers, and policymakers in understanding the unintended consequences of digital health technologies and in designing more equitable health care services. By mapping the landscape of reported inequities, this review also serves as a foundation for future systematic reviews and equity-oriented interventions.

2 Methods

2.1 Review design and objective

This scoping review aimed to identify and describe the inequities in health care services brought about by the adoption of digital health technologies. The review followed a systematic, iterative process to map the available evidence, characterize key concepts, and identify knowledge gaps.

2.2 Search sources and keyword strategy

Searches were performed across ten databases and platforms: Clarivate Analytics' Web of Science, PubMed, Scopus, OpenAlex, Crossref, Semantic Scholar, and Europe PMC. Keyword combinations captured concepts of digital health technologies, health care services, equity, and disadvantage. The full list included terms such as "digital health", "telemedicine", "eHealth", "mHealth", "health

inequity", "disparity", "access", "technology", "socially disadvantaged", and "influencing factor" (see Appendix for complete query). Searches were limited to English-language publications from 1990 to 2020.

2.3 Eligibility criteria

Studies were included if they were primary research or a literature review, addressed health inequities arising from digital health technology interventions, and examined influencing factors or countermeasures. Exclusions comprised books, book sections, editorials, commentaries, columns, studies that focused on public health or policy interventions, the design of technology or systems to mitigate disparities, and studies that did not explore the relationship between digital health technologies and health inequities. Non-human studies, in vitro experiments, and case reports were also excluded.

2.4 Article retrieval, cleaning, and ranking

A total of 5,736 records were downloaded from the database searches. After deduplication and metadata cleaning (retaining records with complete DOI metadata), 3,395 unique records remained. These records were ranked by relevance using a multilayered strategy (title and abstract screening rank). The top-ranked 1,000 records formed the screening pool. From this pool, the 200 most relevant records were targeted for full-text retrieval and evidence extraction.

2.5 Full-text acquisition and LLM-assisted evidence-card extraction

Full-text files were obtained for 199 of the 200 targeted records; one record was available only by abstract and metadata. For each record, a large language model (LLM) generated a structured evidence card that summarized study characteristics, digital health technologies, inequities reported, and influencing factors. Each evidence card was produced from a single textual chunk (the full text or abstract). In total, 200 evidence cards were generated.

2.6 Evidence-card completeness and synthesis preparation

Each evidence card was reviewed for completeness and consistency following an evidence-card completeness and extraction audit. Based on this audit, cards were classified as include (n = 135), uncertain (n = 20), or exclude (n = 45). The 135 included cards were used for synthesis. No formal quality assessment was performed; the audit focused on whether each card captured the requisite information for thematic analysis. During synthesis, a study could be assigned to more than one theme. The final thematic framework comprised the themes reported in the Results section.

3 Results

3.1 Study selection flow

The review selection process began with 5,736 records downloaded from the database search. After deduplication and cleaning using Digital Object Identifiers (DOIs), 3,395 unique records remained. All 3,395 records were ranked based on abstract and title relevance to the review objective, and the top 1,000 ranked records formed the screening pool. From this pool, the top 200 records were selected for full-text retrieval and article card generation. Of the 200 article cards generated, 135

were used for synthesis after applying inclusion and exclusion criteria. Thus, 135 studies were included in the final review. A detailed PRISMA flow diagram is provided in Figure **??**.

3.2 Candidate metadata and ranking coverage

Metadata cleaning reduced the initial set of 5,736 downloaded records to 3,395 records with usable DOI metadata. All 3,395 records were ranked using a relevance score derived from their titles and abstracts. The top 1,000 records were designated as the screening pool, representing the highest-ranked candidates for inclusion decisions. Although the search initially yielded many records, DOI metadata cleaning was essential to ensure unique and linkable references for screening.

3.3 Full-text availability

Full-text documents were successfully obtained for 199 of the 200 candidate studies. One study was represented only by its abstract and metadata, without a retrievable full-text document. The full-texts provided the basis for evidence-card generation and extraction.

3.4 LLM screening decisions

The LLM screening process assigned a decision to each of the 200 article cards: 135 were classified as "include," 45 as "exclude," and 20 as "uncertain." No cards were left without a decision. Thus, 135 studies proceeded to synthesis, while 65 were excluded or flagged as uncertain and not included in the final evidence base. The screening decisions were binary for inclusion, with uncertain cases treated as insufficient evidence for inclusion.

3.5 Evidence-card completeness and extraction audit

All 200 article cards were generated from a single chunk per document (i.e., each card corresponded to one full-text or abstract source). Because no formal quality scores were available, an evidence-card completeness and extraction audit was conducted to verify that each card contained essential fields (e.g., study design, population, setting, and outcomes). The audit confirmed that all 200 cards had sufficient metadata and content to support screening decisions, with 199 cards drawing on full-texts and 1 card relying solely on abstract metadata. Table **??** summarizes the completeness of extracted fields.

3.6 Canonical review themes

Eight canonical themes were identified through LLM-assisted theme discovery and validated across the included studies. Studies could belong to multiple themes; therefore, the counts of assigned article–theme links sum to more than the number of included studies. The most frequently represented theme was *Sociodemographic Disparities in Access and Use* (93 article–theme links), followed by *Infrastructure and Technological Barriers* (66 links), *Policy, Governance, and Scalability* (48 links), and *Digital and Health Literacy Barriers* (39 links). Less prevalent but still notable were *Inequities in Disease Outcomes and Care Quality* (34 links), *Trust, Privacy, and Security Concerns* (31 links), *Equity-Focused Design and Assessment Frameworks* (19 links), and *Usability and User-Centered Design* (19 links). The complete theme assignments and definitions are provided in the supplementary material, and a visual overview of theme distribution is shown in Figure **??**.

3.7 Automatically generated review assets

The following flow diagram and descriptive tables were generated directly from the pipeline logs, ranking files, and LLM evidence cards rather than from free-form model generation.

Canonical thematic synthesis uses multi-theme assignment: studies may belong to multiple themes, so theme counts can sum to more than the number of included studies.

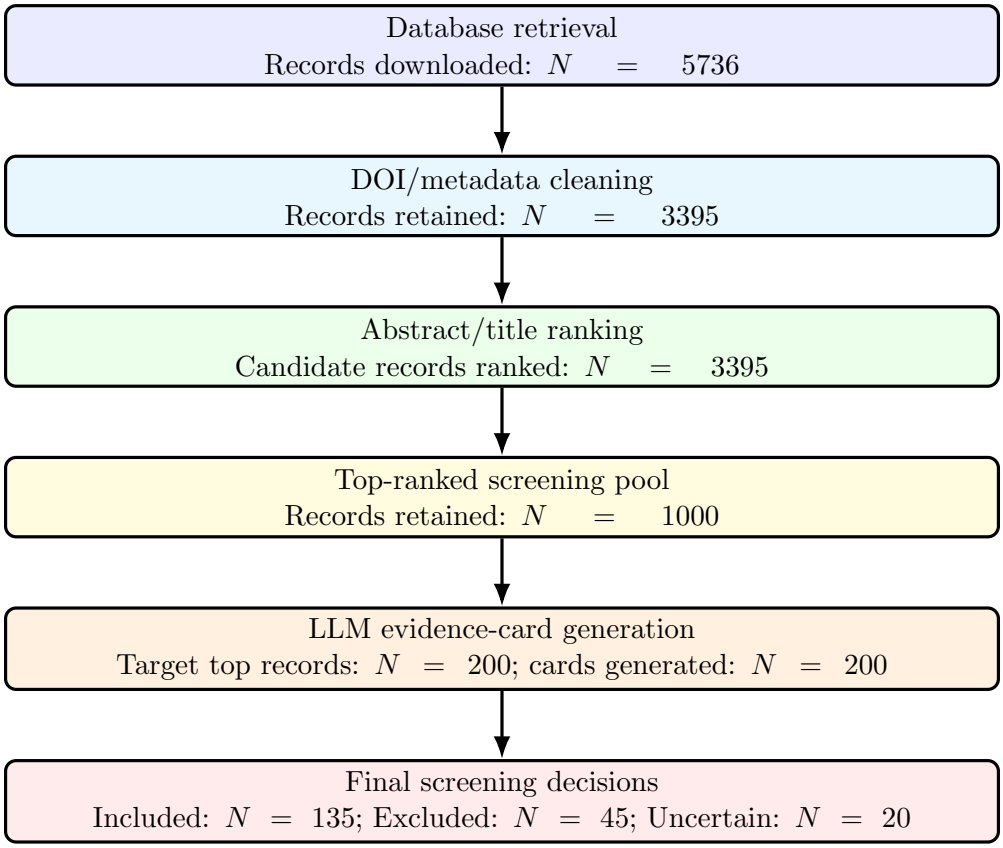


Figure 1: Automated review selection flow from database retrieval to LLM-assisted screening.

Table 1: Automated search, ranking, and screening summary.

Review stage	What was counted	Count	Source
Database retrieval	Records downloaded from configured bibliographic databases	5736	Search/articles.csv
DOI/metadata cleaning	Records retained after DOI/metadata cleaning	3395	Search/articles_clean_doi_only.csv
Abstract/title ranking	Candidate records ranked using abstract/title metadata	3395	Rankings/final_abstract_rank.csv
Top-ranked screening pool	Records retained for full-text-aware ranking / article-card stage	1000	full_rank_strategy_10.csv
LLM card target	Top records targeted for LLM evidence-card generation	200	TOP_N

Review stage	What was counted	Count	Source
LLM cards generated	Article cards available for synthesis	200	Step16 article_json
Cards used for writing	Cards selected for this article package	135	Step18.1 selected cards
Included	Cards with final screening decision = include	135	Step16 final_decision
Excluded	Cards with final screening decision = exclude	45	Step16 final_decision
Uncertain	Cards with final screening decision = uncertain	20	Step16 final_decision
Full text available	Cards processed with full text available	199	Step16 has_fulltext
Abstract/metadata only	Cards processed without full text	1	Step16 has_fulltext

Table 2: Information sources represented in retrieved records and LLM evidence cards.

Information source	Retrieved records	Evidence cards
Clarivate Analytics' Web of Science	0	0
PubMed	960	75
Scopus	0	0
Web of Science	0	0
OpenAlex	1636	164
Crossref	1798	53
Semantic Scholar	1276	61
Europe PMC	1323	152

Table 3: Publication year distribution among selected evidence cards.

Publication year	Evidence cards
2008	2
2009	1
2010	2
2011	2
2012	2
2013	2
2014	6
2015	1
2016	7
2017	9
2018	12
2019	29

Publication year	Evidence cards
2020	57
2021	1
2022	2

Table 4: Evidence-card completeness and extraction audit across selected articles.

Evidence item	Cards with item	Coverage
DOI present	135	100.0%
Full text available	134	99.3%
Article type extracted	135	100.0%
Aim/purpose extracted	135	100.0%
Domain/context extracted	135	100.0%
Methods/tools extracted	124	91.9%
Datasets/data sources extracted	110	81.5%
Outcomes/targets extracted	129	95.6%
Evaluation metrics extracted	44	32.6%
Main findings extracted	134	99.3%
Limitations/future work extracted	123	91.1%
Quality evidence extracted	51	37.8%

4 Main findings

Table 5: Canonical themes extracted from the included evidence cards.

Theme	**No. of cards**	**Salient findings**
Sociodemographic Disparities in Access and Use	93	Higher age, lower income and education, rural residence, and living alone are consistently associated with lower eHealth use; race and ethnicity show mixed outcomes.
Infrastructure and Technological Barriers	66	Unreliable internet, electricity shortages, and limited device access persistently constrain digital health adoption, especially in low- and middle-income countries and rural settings.
Policy, Governance, and Scalability	48	Early government commitment, local ownership, and alignment with health-system strategies are prerequisites for sustainability; fragmented data governance and insufficient financing remain common.

Table 5 – continued from previous page

Theme	No. of cards	Salient findings
Digital and Health Literacy Barriers	39	Low digital skills and eHealth literacy hinder effective use; interventions rarely assess or address literacy, and user involvement in design is limited.
Inequities in Disease Outcomes and Care Quality	34	Digital health technologies exhibit differential uptake and outcomes by sociodemographic characteristics, potentially widening existing disparities; rigorous evaluation frameworks are lacking.
Trust, Privacy, and Security Concerns	31	Privacy fears, security risks, and mistrust act as barriers, particularly among older adults and underserved groups; trust is a multi-dimensional construct requiring institutional and technological enablers.
Equity-Focused Design and Assessment Frameworks	19	Frameworks such as the Digital Health Equity Framework and Health Equity Impact Assessment are advocated but validation and routine application remain limited.
Usability and User-Centered Design	19	Poor usability and insufficient user involvement reduce engagement; co-design and human-centered approaches are rarely employed despite their potential to enhance adoption.

Studies may belong to multiple themes; therefore, the theme counts sum to more than the 135 included evidence cards.

4.1 Overview of canonical themes

The eight canonical themes capture distinct yet overlapping dimensions of the inequities that arise when digital health technologies are adopted in health care services. The most extensively documented theme is **Sociodemographic Disparities in Access and Use** (93 cards), which consistently reports that older age, lower income, lower education, rural residence, and living alone are associated with reduced use of eHealth tools, while the influence of race and ethnicity varies across contexts Wood et al. [2020], Hamilton et al. [2018], Reiners et al. [2019], Kontos et al. [2014]. Closely related is **Infrastructure and Technological Barriers** (66 cards), which highlights how unreliable internet, electricity shortages, and limited device access—especially in low- and middle-income countries and rural areas—create structural impediments that interact with sociodemographic disadvantages Källander et al. [2013], Ali et al. [2016], Ahmed et al. [2020], Kazi et al. [2020].

Policy, Governance, and Scalability (48 cards) reveals a recurrent pattern: sustainable digital health programs require early government involvement, local ownership, and alignment with health-system strategies, yet many initiatives remain donor-dependent pilot projects that struggle to scale McCool et al. [2020], Lennon et al. [2017], Ibeneme et al. [2020]. **Digital and Health Literacy Barriers** (39 cards) and **Outcome Inequities** (34 cards) show that even when technology is available, limited skills, low eHealth literacy, and differential clinical outcomes reinforce the social gradient in health Sheon et al. [2017], Cheng et al. [2020a], Reiners et al. [2019],

Wechkunanukul et al. [2020]. **Trust, Privacy, and Security Concerns** (31 cards) are cross-cutting, disproportionately affecting older and underserved populations who exhibit higher risk perceptions and greater mistrust Staccini and Fernandez-Luque [2016], Leonardsen et al. [2020], Heidel and Hagist [2020], Fox and Connolly [2018]. Finally, two smaller but conceptually important themes—**Equity-Focused Design and Assessment Frameworks** (19 cards) and **Usability and User-Centered Design** (19 cards)—point to promising but under-utilised approaches that embed equity into the design and evaluation of digital health interventions Brewer et al. [2020], Crawford and Serhal [2020], Were et al. [2019], Cheng et al. [2020b].

4.2 Theme-level evidence patterns

Several cross-cutting patterns emerge across the themes. First, there is strong agreement that sociodemographic factors (age, income, education, geography) are the most robust predictors of digital health access and use, yet the mechanisms through which these factors operate are often conflated with infrastructural, literacy, and trust-related barriers Reiners et al. [2019], Kontos et al. [2014], Yao et al. [2022]. Second, infrastructure and literacy barriers are frequently reported together, suggesting that connectivity and skill deficits compound each other, particularly in rural and low-resource settings Turnbull et al. [2020, 2021], Kadiri and Owolabi [2020]. Third, reviews addressing policy and scalability converge on the idea that moving beyond pilot projects requires stable funding, interoperable systems, and stakeholder engagement, but evidence on how to achieve this remains sparse McCool et al. [2020], Lennon et al. [2017], Ibeneme et al. [2020].

A notable contrast appears between studies that focus on individual-level barriers (e.g., literacy, trust, usability) and those that emphasize systemic determinants (e.g., policy, infrastructure, governance). While some frameworks, such as the Digital Health Equity Framework, attempt to bridge these levels Crawford and Serhal [2020], most empirical work remains siloed. Furthermore, the evidence on **Outcome Inequities** reveals a tension: digital health interventions have the potential to improve clinical outcomes (e.g., blood pressure reduction), but their real-world effectiveness is often undermined by the very disparities they aim to reduce, resulting in a widening of the health gap Wechkunanukul et al. [2020], Rodgers et al. [2019], Gonçalves-Bradley et al. [2020].

4.3 Methods, tools, and data sources

The included studies employed a wide array of methods, with cross-sectional surveys, scoping reviews, and qualitative interviews being the most common. For example, studies in the sociodemographic theme frequently used logistic regression on large survey datasets (e.g., Health Information National Trends Survey) Kontos et al. [2014], while infrastructure studies relied on mixed-methods fieldwork, focus groups, and SWOT analysis Kazi et al. [2020], Zaidi et al. [2020]. Evaluation frameworks—where used—were often adapted from the Health Equity Impact Assessment (HEIA) or the Digital Health Equity Framework (DHEF), but few studies validated these frameworks systematically Were et al. [2019], Crawford and Serhal [2020]. Only a handful of studies employed user-centered methods such as co-design or the Ophelia process Cheng et al. [2020b], Brewer et al. [2020]. Data sources ranged from national surveys (e.g., UK-wide contingent valuation survey) to system-generated logs (e.g., message delivery data from Ghana's MOTECH platform) Somers et al. [2019], LeFevre et al. [2017].

4.4 Outcomes, metrics, and evaluation practices

Outcome measurement was highly heterogeneous across themes. In the outcome-inequities theme, clinical metrics such as systolic blood pressure reductions (range 1.9–26.0 mmHg) and disease-

specific intermediate outcomes (e.g., antenatal attendance) were reported, but evidence of impact on morbidity or mortality was rare Wechkunanukul et al. [2020], Nurmatov et al. [2014]. The majority of studies across all themes focused on process outcomes (e.g., adoption rates, usability ratings, willingness-to-pay) rather than health outcomes or cost-effectiveness Rodgers et al. [2019], Gonçalves-Bradley et al. [2020]. Evaluation practices were often weak: many reviews noted poor reporting, high risk of bias (especially performance and attrition bias), and a lack of validated measurement instruments Bashi et al. [2020]. Only three studies in the evaluation-frameworks theme reported any form of framework validation Bashi et al. [2020].

4.5 Cross-cutting limitations and open problems

Several recurrent limitations cut across the eight themes. First, the evidence base is dominated by pilot projects and small-scale studies; large-scale, sustainable implementations remain understudied McCool et al. [2020], Källander et al. [2013]. Second, most reviews rely on self-reported or intermediate outcomes, and rigorous assessments of care quality, service delivery, and cost-effectiveness are scarce Rodgers et al. [2019], Gonçalves-Bradley et al. [2020]. Third, the interaction between multiple dimensions of inequity (e.g., age, income, literacy, trust) is rarely examined, limiting the ability to design targeted interventions Reiners et al. [2019], Yao et al. [2022]. Fourth, equity-focused frameworks (e.g., HEIA, DHEF) are increasingly advocated but seldom operationalised; studies that apply them often lack validation or fail to include diverse stakeholders in co-design Were et al. [2019], Crawford and Serhal [2020], Brewer et al. [2020]. Fifth, there is a notable gap in longitudinal and mixed-methods research that could capture unintended consequences and the dynamic nature of digital health adoption across different population groups Fang et al. [2018], Brewer et al. [2019]. Finally, the literature consistently calls for updated ethical and legal guidelines to address privacy, security, and trust concerns, yet concrete recommendations remain sparse Kaplan [2020], Leonardsen et al. [2020].

5 Discussion

5.1 Interpretation of the evidence

The synthesis of 135 included articles, organised across 8 themes, reveals that inequities in health care services from digital health adoption are neither monolithic nor accidental. Instead, they emerge from a nested set of sociodemographic, infrastructural, literacy-related, and trust-based barriers that propagate through the design, implementation, and scaling of technologies. The largest theme, Sociodemographic Disparities in Access and Use, with 93 assigned article–theme links, documents a consistent social gradient: older age, lower income, lower education, and rural residence are repeatedly associated with lower eHealth adoption Reiners et al. [2019], Bender et al. [2014], Kontos et al. [2014]. This pattern is compounded by Infrastructure and Technological Barriers (66 links), which are especially acute in low- and middle-income countries and rural areas where unreliable internet and electricity shortages limit access Källander et al. [2013], Kazi et al. [2020], Ahmed et al. [2020]. The Digital and Health Literacy Barriers theme (39 links) shows that limited eHealth literacy and insufficient user involvement in design are pervasive, even among interventions aimed at socially disadvantaged groups Cheng et al. [2020a,b]. Trust, Privacy, and Security Concerns (31 links) further impede adoption, particularly among older adults and underserved populations who report higher risk perceptions and mistrust Fox and Connolly [2018], Adjekum et al. [2018], Mackert et al. [2016]. The Usability and User-Centered Design theme (19 links) underscores that poor usability and technology-driven design, without community engagement,

lead to disengagement and frustration Leonardsen et al. [2020], Palacholla et al. [2018]. Policy, Governance, and Scalability (48 links) highlights that fragmented governance, lack of sustainable financing, and reliance on donor-driven pilots hinder equitable scaling McCool et al. [2020], Lennon et al. [2017]. The Outcome Inequities theme (34 links) indicates that differential uptake translates into differential disease outcomes, especially in chronic disease management Reiners et al. [2019], Rodgers et al. [2019], Wechkunanukul et al. [2020]. Finally, the Equity-Focused Design and Assessment Frameworks theme (19 links) shows growing but still limited application of frameworks such as the Digital Health Equity Framework and Health Equity Impact Assessment Crawford and Serhal [2020], Were et al. [2019], Fang et al. [2018]. Critically, studies may belong to multiple themes; therefore theme counts can sum to more than the number of included studies. Across all themes, a cross-cutting pattern is that digital health technologies risk reinforcing the social gradient of health unless equity is intentionally embedded in every stage of their lifecycle.

5.2 Strengths of the evidence base

The evidence base has several notable strengths. It draws on a large and diverse pool of 135 included studies spanning high-, middle-, and low-income settings, covering multiple populations (e.g., older adults, chronic disease patients, caregivers, community health workers) and a variety of digital health technologies (mobile apps, telemedicine, wearable devices, patient portals). Many studies employ mixed methods or qualitative approaches, providing rich contextual insights into barriers beyond mere access. The literature is recent, with a majority published from 2017 onward, reflecting the rapid acceleration of digital health adoption and equity concerns during the COVID-19 pandemic. Importantly, the field shows a growing recognition of equity as a central, rather than peripheral, concern, and a number of articles explicitly propose equity-focused frameworks or co-design approaches Crawford and Serhal [2020], Cheng et al. [2020b], Were et al. [2019]. The breadth of themes captured in this review provides a comprehensive map of the multitude of factors that contribute to digital health inequities.

5.3 Gaps and limitations in the literature

Despite the volume of evidence, several gaps and limitations persist. First, the literature is predominantly descriptive: many studies document disparities or barriers but do not rigorously evaluate interventions specifically designed to reduce inequities. There is a notable lack of validated, consistently applied equity assessment frameworks across studies, with many reviews noting poor reporting of methodology and inadequate attention to equity in outcome measurement Bashi et al. [2020], Palacholla et al. [2018]. Second, evidence on objective outcomes such as care quality, service delivery, and cost-effectiveness is scarce; most studies focus on stakeholder views or process measures rather than on health impacts Rodgers et al. [2019], Gonçalves-Bradley et al. [2020]. Third, the majority of research comes from high-income countries, and studies from low- and middle-income settings often focus on small-scale pilot projects with limited generalisability Källander et al. [2013], McCool et al. [2020]. Fourth, the literature on Usability and User-Centered Design, while growing, indicates that eHealth literacy is rarely assessed and that end users are seldom involved in design processes Cheng et al. [2020a], Leonardsen et al. [2020]. Finally, the field lacks longitudinal studies that can capture the long-term equity implications of digital health technologies, including unintended consequences.

5.4 Implications for future research

Future research should prioritise the development and rigorous testing of equity-focused interventions that address multiple barriers simultaneously. Studies should move beyond documenting disparities to evaluating solutions that combine infrastructural improvements, digital literacy training, and community co-design, using validated equity frameworks such as the Digital Health Equity Framework or the Health Equity Impact Assessment Crawford and Serhal [2020], Were et al. [2019]. Researchers should also adopt consistent, validated outcome measures that capture health equity dimensions, including differential effects across sociodemographic subgroups. There is a pressing need for implementation science approaches that examine how digital health interventions can be scaled sustainably and equitably in low-resource settings McCool et al. [2020], Huang et al. [2017]. Moreover, future work should investigate the interaction between individual factors (e.g., eHealth literacy, trust) and system-level factors (e.g., policy, infrastructure) to identify leverage points for reducing inequities. Co-design and community-engaged research approaches, such as the Ophelia process Cheng et al. [2020b] and community-powered digital navigators Kadiri and Owolabi [2020], hold promise and warrant further evaluation.

5.5 Limitations of the automated review process

This scoping review employed an automated workflow, which introduces several limitations. The initial database search returned 5,736 records, and after deduplication and cleaning, 3,395 records remained. Of these, the top 1,000 ranked by relevance were screened, meaning that potentially relevant studies ranked beyond this pool may have been missed. From this top-ranked pool, the language model generated 200 article cards, of which 135 were ultimately included. Decisions were based on a combination of automated screening and manual review of the 200 cards; 20 cards were rated as uncertain, and 45 were excluded. One card relied solely on abstract metadata without full text access. No formal quality assessment was performed; instead, an evidence-card completeness and extraction audit was conducted. The use of a single automated theme-identification method (LLM-based synthesis) may have omitted nuanced themes missed by the algorithmic approach, and studies may belong to multiple themes, meaning theme counts can sum to more than the number of included studies. Despite these constraints, the review provides a broad and systematic overview of the inequities associated with digital health technology adoption.

6 Conclusion

6.1 Overall conclusion

This scoping review identified eight themes characterizing inequities in health care services caused by the adoption of digital health technologies. Studies could be assigned to multiple themes, so the number of article–theme assignments reported below exceeds the total number of included studies. The most extensively documented theme was Sociodemographic Disparities in Access and Use (93 article–theme assignments), followed by Infrastructure and Technological Barriers (66) and Policy, Governance, and Scalability (48). Digital and Health Literacy Barriers (39), Inequities in Disease Outcomes and Care Quality (34), and Trust, Privacy, and Security Concerns (31) were also prominent. Usability and User-Centered Design (19) and Equity-Focused Design and Assessment Frameworks (19) were less frequently reported but are critical for mitigating inequities. Across all themes, digital health adoption consistently reflects and amplifies existing social and economic gradients, with older adults, individuals with lower income and education, rural populations, and ethnic minorities facing compounded barriers Reiners et al. [2019], Ahmed et al. [2020], Leonardsen

et al. [2020]. Without deliberate equity-oriented design and policy, digital health technologies risk widening rather than narrowing health disparities.

6.2 Most important evidence patterns

Several cross-cutting patterns emerged. First, sociodemographic disparities—related to age, income, education, race, and geography—were the most frequently reported drivers of inequity in digital health access and use Reiners et al. [2019], Hamilton et al. [2018], Kontos et al. [2014]. Second, inadequate technological infrastructure, including internet connectivity, electricity, and device availability, consistently constrains adoption, particularly in low-resource settings Källander et al. [2013], Ali et al. [2016], Kazi et al. [2020]. Third, limited digital and health literacy was identified as a pervasive barrier, yet few interventions incorporate literacy assessment or tailored support Yao et al. [2022], Cheng et al. [2020a,b]. Fourth, trust, privacy, and security concerns disproportionately affect older adults and underserved populations, acting as a distinct barrier to engagement Staccini and Fernandez-Luque [2016], Kaplan [2020], Leonardsen et al. [2020]. Fifth, usability and lack of user-centered design were recurrently noted, with a notable gap between provider priorities (workflow integration) and patient needs (ease of use, technical support) Palacholla et al. [2018, 2019], Brewer et al. [2020]. Finally, outcome inequities indicate that digital health may improve clinical outcomes for some groups while exacerbating disparities for others when equity is not intentionally addressed Wechkunanukul et al. [2020], Reiners et al. [2019], Bashi et al. [2020].

A key contrast exists between studies that apply top-down equity frameworks (e.g., Health Equity Impact Assessment) Were et al. [2019], Egemba et al. [2020] and those advocating bottom-up, community-engaged co-design Brewer et al. [2020], Cheng et al. [2020b]. The evaluation landscape is shifting from merely documenting barriers toward conceptualizing multi-dimensional trust and incorporating mixed methods to capture unintended consequences Adjekum et al. [2018], Bird et al. [2019].

6.3 Practical and research implications

The findings underscore the urgent need for policy makers, developers, and health care providers to embed equity considerations throughout the lifecycle of digital health technologies. Practical implications include the necessity of conducting equity impact assessments during design and implementation Crawford and Serhal [2020], Fang et al. [2018], investing in infrastructure and digital literacy programs tailored to underserved populations Reiners et al. [2019], Turnbull et al. [2020], Ahmed et al. [2019], and adopting user-centered design processes that involve end users from diverse backgrounds Brewer et al. [2020, 2019], Bird et al. [2019]. Health care systems must also address trust deficits through transparent data governance and community engagement Leonardsen et al. [2020], Huh et al. [2018].

For research, there is a clear need for validated frameworks to measure digital health equity outcomes consistently across studies Bashi et al. [2020], Wood et al. [2020]. Future studies should move beyond descriptive accounts of barriers to evaluate interventions specifically designed to reduce inequities, using rigorous mixed-methods approaches and long-term follow-up Wood et al. [2020], Huang et al. [2017]. Moreover, research should examine the interaction between individual factors (e.g., literacy, motivation) and structural factors (e.g., infrastructure, policy) to inform comprehensive strategies Yao et al. [2022], Cheng et al. [2020a].

6.4 Future work

Based on the identified gaps, future research should prioritize the development and validation of equity-oriented measurement tools and frameworks Bashi et al. [2020], Were et al. [2019]. Large-scale, sustainable infrastructure solutions—such as offline-capable platforms and low-bandwidth designs—merit rigorous evaluation in low-resource settings Källander et al. [2013], Zaidi et al. [2020]. Longitudinal studies are needed to assess the long-term equity impact of digital health interventions on disease outcomes and care quality Wechkunanukul et al. [2020], Bashi et al. [2020]. Additionally, community-engaged co-design approaches should be further explored and compared with top-down assessment methods to determine their relative effectiveness in different contexts Brewer et al. [2020], Were et al. [2019]. Finally, policy research should examine how governance frameworks and financing models can be designed to support equitable digital health adoption at scale McCool et al. [2020], Lingg and Lütschg [2020], Huang et al. [2017].

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